

Artificial Intelligence Exam Notes

CT 710 (BEI, IV-I) · CT 653 (BCT, III-II) · CT 78506 (BEX, IV-II) · TU, IOE · tiers over 41 papers, 2069–2082 BS

2. Problem Solving – Solved Problems

Every worked problem here opens with the exam question **exactly as the paper prints it**, typos and all, so it can be attempted from this document alone. Methods follow *Insights on Artificial Intelligence*; where that book is wrong or skips a step, the notes say so. · Every answer in this file is machine-checked by `verify.py`.

The five things everything here is built on:

- **1. A crypt-arithmetic column is an equation.** $(\text{column sum}) + C_{\text{in}} = 10 C_{\text{out}} + (\text{result digit})$. Every deduction in §2.N.1 comes from one of these.
- **2. A carry is 0 or 1.** Adding two digits plus a carry gives at most $9 + 9 + 1 = 19$. With *three* addends it can reach $9 + 9 + 9 + 1 = 28$, so a carry of **2** is possible in TEN + TEN + FORTY. Nothing else in these papers has three addends.
- **3. If the answer is one digit longer than the addends, that leading digit is 1** – two 4-digit numbers cannot reach 20000. This single line gives $M = 1$ in SEND + MORE, $G = 1$ in BASE + BALL, $P = 1$ in LOGIC + LOGIC and $D = 1$ in CROSS + ROADS, for free.
- **4. $2X$ ends in X only when $X = 0$.** ($X = 5$ gives 10, whose last digit is 0, not 5.) This cracks TEN + TEN + FORTY and LOVE + LOVE.
- **5. A jug state is the ordered pair of contents.** Every water-jug operator maps (x, y) to another (x, y) ; there is nothing else to track.

Read this before the crypt-arithmetic problems

- **Seven of the thirteen puzzles set in these papers have more than one valid answer.** ONE + ONE + TWO = FOUR has **107**. RIGHT + RIGHT = WRONG, WRONG + WRONG = RIGHT and ONE + ONE = TWO pin **no letter at all**.
- So the marks are in the **column reasoning**, not the digits. Write X, D, C and the carries, then argue. Where the constraints run out, say “the constraints leave N free; take $N = 3$ ” and carry on. That sentence is worth more than the answer.
- Uniqueness of each puzzle is stated below it, counted by exhaustive search in `verify.py` over every letter-to-digit assignment.

2.N.1 Crypt-arithmetic: the six puzzles with a unique answer

SEND + MORE = MONEY 1+6 8

HOT 5/41 72 Ash 1+6, 76 Ba 8, 74 Ma 8, 74 Bh 8, 72 Ash 8 · **unique solution** · the classic, and the one *Insights* works in full

QUESTION AS PRINTED 72 Ash · Q2 · 1+6

Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps. Different letters denote different integers and identical letters denote the same integer.

$$\begin{array}{r} \text{S E N D} \\ + \text{M O R E} \\ \hline \text{M O N E Y} \end{array}$$

Formulation

- $X = \{S, E, N, D, M, O, R, Y\}$ (8 letters)
- $D_{\text{om}} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- C :
 - $S \neq 0, M \neq 0$ (leading letters)
 - all eight letters take different digits
 - $D + E = 10C_1 + Y$
 - $N + R + C_1 = 10C_2 + E$
 - $E + O + C_2 = 10C_3 + N$
 - $S + M + C_3 = 10C_4 + O$
 - $C_4 = M$

$$\begin{array}{r}
 \begin{array}{cccc}
 C_4 & C_3 & C_2 & C_1 \\
 S & E & N & D \\
 + & M & O & R & E \\
 \hline
 M & O & N & E & Y
 \end{array}
 \end{array}$$

The deduction chain

- $M = 1$. $C_4 = M$, and a carry is at most 1. $M \neq 0$ because it leads MONEY, so $M = 1$.
- $O = 0$. Column 4 is $S + 1 + C_3 = 10 \cdot 1 + O$. The largest $S + 1 + C_3$ can be is $9 + 1 + 1 = 11$, so O is 0 or 1; 1 is taken by M , so $O = 0$.
- $S = 9$. With $O = 0$, column 4 reads $S + 1 + C_3 = 10$. Column 3 is $E + 0 + C_2 = 10C_3 + N$, and $E + C_2 \leq 10$, so $C_3 = 0$ unless $E + C_2 = 10$, which would force $N = 0 = O$. So $C_3 = 0$ and $S = 9$.
- $N = E + 1$ and $C_2 = 1$. Column 3 now reads $E + C_2 = N$. If C_2 were 0 then $E = N$, not allowed. So $C_2 = 1$ and $N = E + 1$.
- $R = 8$. Column 2 is $N + R + C_1 = 10 + E$. Substitute $N = E + 1$: $E + 1 + R + C_1 = E + 10$, so $R + C_1 = 9$. $R = 9$ is taken by S , so $C_1 = 1$ and $R = 8$.
- $D = 7$. Column 1 is $D + E = 10 + Y$ (since $C_1 = 1$), so $D + E \geq 12$. With 9 and 8 gone, the only pairs are $5 + 7$ and $6 + 7$, so one of D, E is 7. If $E = 7$ then $N = 8 = R$, so $D = 7$, and E is 5 or 6.
- $E = 5, N = 6, Y = 2$. $E = 6$ would give $N = 7 = D$. So $E = 5, N = 6$, and $Y = D + E - 10 = 2$.

Answer: $S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2 \rightarrow 9567 + 1085 = 10652$. Unique: exactly one assignment satisfies the constraints.

BASE + BALL = GAMES 1+6 2+6 3+5 8

HOT 7/41 75 Ba 2+6, 82 Ka 2+6, 82 Ba 2+6, 79 Ash 2+6, 82 Bh 3+5, 79 Jth 1+6, 73 Bh 8 · the most-asked puzzle in the subject · unique solution

QUESTION AS PRINTED 79 Ash · 2+6 · also 75 Ba, 82 Ka, 82 Ba, 82 Bh, 79 Jth, 73 Bh

What do you understand by Constraint Satisfaction Problem? Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps. Different letters denote different integers and identical letters denote the same integer.

$$\begin{array}{r}
 B A S E \\
 + B A L L \\
 \hline
 G A M E S
 \end{array}$$

Formulation

- $X = \{B, A, S, E, L, G, M\}$ (7 letters)
- $D_{\text{om}} = \{0, \dots, 9\}$
- C :
 - $B \neq 0, G \neq 0$
 - all seven letters differ
 - $E + L = 10C_1 + S$
 - $S + L + C_1 = 10C_2 + E$
 - $A + A + C_2 = 10C_3 + M$
 - $B + B + C_3 = 10C_4 + A$
 - $C_4 = G$

$$\begin{array}{r}
 \begin{array}{cccc}
 C_4 & C_3 & C_2 & C_1 \\
 & B & A & S & E \\
 + & & B & A & L & L \\
 \hline
 & G & A & M & E & S
 \end{array}
 \end{array}$$

Answer: $B=7, A=4, S=8, E=3, L=5, G=1, M=9 \rightarrow 7483 + 7455 = 14938$. Unique.

TEN + TEN + FORTY = SIXTY 2+6

PIN 3/41 80 Bh 2+6, 78 Ba 2+6, 78 Po 2+6 (printed FORTY + TEN + TEN, same puzzle) · the only puzzle with three addends, so here a carry can be 2 · unique solution

QUESTION AS PRINTED 80 Bh · 2+6 · 78 Ba; 78 Po prints the addends as FORTY + TEN + TEN

What do you understand by Constraint Satisfaction Problem? Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps.

$$\begin{array}{r}
 \text{T E N} \\
 + \quad \text{T E N} \\
 + \text{F O R T Y} \\
 \hline
 \text{S I X T Y}
 \end{array}$$

The deduction chain

- $G = 1$. Two 4-digit numbers cannot reach 20000, so the leading carry $C_4 = G = 1$.
- $L = 5, C_1 = 0, C_2 = 1$. This is the clever step. Add the column-1 and column-2 equations:

$$(E + L) + (S + L + C_1) = (10C_1 + S) + (10C_2 + E)$$
 Everything cancels except $2L = 9C_1 + 10C_2$. With $2L \leq 18$ the only possibilities are $(C_1=0, C_2=0, L=0)$ and $(C_1=0, C_2=1, L=5)$; $2L = 9$ and $2L = 19$ are odd, so no other case survives. $L = 0$ makes column 1 read $E = S$, which the AllDiff constraint forbids. So $L = 5$.
- $S = E + 5$. Column 1 with $L = 5, C_1 = 0$ is $E + 5 = S$. Hence $E \leq 4$, and $E \neq 0$ (else $S = 5 = L$), so $E \in \{2, 3, 4\}$ and $S \in \{7, 8, 9\}$.
- **Columns 3 and 4.** $2A + 1 = 10C_3 + M$ and $2B + C_3 = A + 10$.
- **Test the three cases for E :**
 - $E = 2, S = 7$: $C_3=0$ forces $M = 9, A = 4$, then $2B = 14, B = 7 = S$. Clash. $C_3=1$ gives no legal B . ×
 - $E = 4, S = 9$: every branch needs $2B$ odd, or repeats G . ×
 - $E = 3, S = 8$: $C_3 = 0, M = 2A + 1$ odd and unused $\rightarrow M = 9, A = 4$; then $2B + 0 = 14$, so **B = 7**. ✓

EAT + THAT = APPLE 2+5 2+6

PIN 2/41 80 Ba 2+6, 80 Ch 2+5 · the only puzzle whose addends differ in length · unique solution

QUESTION AS PRINTED 80 Ba · 2+6 · also 80 Ch 2+5

What do you understand by Constraint Satisfaction Problem? Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps.

$$\begin{array}{r} \text{E A T} \\ + \text{T H A T} \\ \hline \text{A P P L E} \end{array}$$

Formulation

- $X = \{E, A, T, H, P, L\}$; $E, T, A \neq 0$; all six differ.
- Columns right to left:
 - $T + T = 10C_1 + E$
 - $A + A + C_1 = 10C_2 + L$
 - $E + H + C_2 = 10C_3 + P$
 - $T + C_3 = 10C_4 + P$
 - $C_4 = A$

$$\begin{array}{r} C_3 C_1 \\ \text{E A T} \\ + \text{T H A T} \\ \hline \text{A P P L E} \end{array}$$

The deduction chain

- $A = 1$. A 3-digit plus a 4-digit number is at most $999 + 9999 = 10998$, so the 5-digit answer begins with 1.
- $T = 9$. The sum is at least 10000, so $\text{THAT} \geq 10000 - 999 = 9001$, forcing $T = 9$.
- $E = 8, C_1 = 1$. Column 1 is $9 + 9 = 18$, so $E = 8$ with a carry of 1.
- $L = 3, C_2 = 0$. Column 2 is $1 + 1 + 1 = 3$, so $L = 3$, no carry.
- $P = 0, C_3 = 1$. Column 4 is $T + C_3 = 10C_4 + P$ with $C_4 = A = 1$, i.e. $9 + C_3 = 10 + P$, so $P = C_3 - 1$. A carry is 0 or 1, so $C_3 = 1$ and $P = 0$.
- $H = 2$. Column 3 is $E + H + C_2 = 10C_3 + P$, i.e. $8 + H + 0 = 10 + 0$, so $H = 2$.
- **Every letter fell out of the two length observations.** No case-splitting was needed at all, which is why this puzzle is worth doing first if you are short of time.

Answer: $E=8, A=1, T=9, H=2, P=0, L=3 \rightarrow 819 + 9219 = 10038$. Unique.

CROSS + ROADS = DANGER 2+6

PIN 1/41 81 Bh 2+6 · unique solution

QUESTION AS PRINTED 81 Bh · 2+6

What do you understand by Constraint Satisfaction Problem? Discuss it, and list all the constraints. Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps.

$$\begin{array}{r} \text{C R O S S} \\ + \text{R O A D S} \\ \hline \text{D A N G E R} \end{array}$$

Formulation

- $X = \{C, R, O, S, A, D, N, G, E\}$ (9 letters, so one digit goes unused)
- $C, R, D \neq 0$; all nine differ.
- Columns right to left:
 - $S + S = 10a_1 + R$
 - $S + D + a_1 = 10a_2 + E$
 - $O + A + a_2 = 10a_3 + G$
 - $R + O + a_3 = 10a_4 + N$
 - $C + R + a_4 = 10a_5 + A$
 - $a_5 = D$

The deduction chain

- $D = 1$. Two 5-digit numbers cannot reach 200000, so the leading carry $a_5 = D = 1$.
- R is even. R is the last digit of $2S$, and $2S$ is even.
- $C + R + a_4 = A + 10$, from the leftmost column with $a_5 = 1$. Since $C, R \leq 9$, this needs C and R both large.
- These three plus AllDiff leave a small search. Propagating them column by column (the constraint-propagation loop of §2.5) terminates on exactly one assignment, below.
- **The column check, which is the part that earns the marks:**
 - col 1: $3 + 3 = 6 = R, a_1 = 0 \checkmark$
 - col 2: $3 + 1 + 0 = 4 = E, a_2 = 0 \checkmark$
 - col 3: $2 + 5 + 0 = 7 = G, a_3 = 0 \checkmark$
 - col 4: $6 + 2 + 0 = 8 = N, a_4 = 0 \checkmark$
 - col 5: $9 + 6 + 0 = 15 = A + 10 = 5 + 10, a_5 = 1 = D \checkmark$
- Digit 0 is the one left unused.

Answer: $C=9, R=6, O=2, S=3, A=5, D=1, N=8, G=7, E=4 \rightarrow 96233 + 62513 = 158746$. Unique.

2.N.2 Crypt-arithmetic: the seven puzzles with more than one answer

These are set just as often as the unique ones, and a marker cannot be expecting a particular set of digits. **Show the forced letters, name the free ones, choose values, and verify.** The counts below are from exhaustive search in `verify.py`.

Puzzle	Answers	Forced by the constraints	A valid assignment, and why the rest is free
ONE + ONE + TWO = FOUR 71 Ma, 73 Bh, 80 Ash	107	$F = 1$ only	$O=6, N=4, E=2, T=3, W=8, F=1, U=7, R=0$: $642 + 642 + 386 = 1670$. Eight letters, but only the leading-carry constraint bites.
LOVE + LOVE = HATE 81 Ba	42	$E = 0$ only	$L=3, O=5, V=6, E=0, H=7, A=1, T=2$: $3560 + 3560 = 7120$. $E = 0$ because $2E$ must end in E (rule 4).
WRONG + WRONG = RIGHT 70 Bh, 77 Ch	21	nothing	$W=3, R=7, O=0, N=8, G=1, I=4, H=6, T=2$: $37081 + 37081 = 74162$.
SWIM + WEAR = RELAX 74 Bh, 81 Ch	16	$R = 1, I = 0$	$S=9, W=4, I=0, M=5, E=3, A=2, R=1, L=7, X=6$: $9405 + 4321 = 13726$. $R = 1$ is the leading carry.
ONE + ONE = TWO Insights §2.3	16	nothing	$O=2, N=3, E=1, T=4, W=6$: $231 + 231 = 462$. This is the book's own worked example, and it presents its answer as though it were the only one.
RIGHT + RIGHT = WRONG 69 Po	11	nothing	$R=4, I=2, G=0, H=6, T=5, W=8, O=1, N=3$: $42065 + 42065 = 84130$.
TWO + TWO = FOUR 78 Ch	7	$F = 1$ only	$T=7, W=3, O=4, F=1, U=6, R=8$: $734 + 734 = 1468$.

How to write one of these up for full marks

- 1. Give X , D and C with the carry equations, exactly as for a unique puzzle. **This is most of the mark scheme.**
- 2. Derive whatever is forced. There is always something: a leading carry, or a $2X$ ending in X .
- 3. Say plainly that the constraints do not determine the rest, then pick values that satisfy every column and the AllDiff rule.
- 4. **Verify by doing the addition.** One line, and it converts a guess into a demonstrated answer.

Worked example of step 2: LOVE + LOVE = HATE

- Column 1: $E + E = 10C_1 + E \Rightarrow E = 10C_1$. A digit, so $E = 0, C_1 = 0$.
- Column 2: $2V + 0 = 10C_2 + T$, so T is even.
- Column 3: $2O + C_2 = 10C_3 + A$.
- Column 4: $2L + C_3 = H$, and $H \leq 9$ forces $L \leq 4$.
- That is everything the constraints give. $L \in \{1, 2, 3, 4\}$ and the rest follows the choice. Take $L = 3, O = 5, V = 6$: then $T = 2, A = 1, H = 7$, and $3560 + 3560 = 7120$ ✓

AB + CD = AAA: what values can B take? 4

76 Bh 4 · **not a solve-it question**: it asks what is *possible*, so the answer is a set and a justification

QUESTION AS PRINTED 76 Bh · 4

In a correctly worked out crypt-arithmetic problem $AB + CD = AAA$, what could be the possible values of B? Justify your answer.

Reasoning

- $A = 1$. Two 2-digit numbers sum to at most $99 + 99 = 198$. The only 3-digit repdigit AAA in range is **111**, so $A = 1$.
- So $AB = 10 + B$ and $CD = 111 - (10 + B) = 101 - B$.
- $B \geq 3$. CD must be a 2-digit number, so $101 - B \leq 99$, giving $B \geq 2$. And $B = 2$ makes $CD = 99$, i.e. $C = D = 9$, which the AllDiff rule forbids. So $B \geq 3$.
- $B \leq 8$. For B from 3 to 9, $CD = 101 - B$ runs $98, 97, \dots, 92$, so $C = 9$ throughout. $B = 9$ would make $B = C = 9$. **This is the case that is easy to miss**, and it is the whole difficulty of the question.
- $B = 1$ is out because $A = 1$, and $B = 0$ would need $CD = 101$.

Answer: $B \in \{3, 4, 5, 6, 7, 8\}$, six values

B	AB	CD	Check
3	13	98	$13 + 98 = 111$ ✓
4	14	97	$14 + 97 = 111$ ✓
5	15	96	$15 + 96 = 111$ ✓
6	16	95	$16 + 95 = 111$ ✓
7	17	94	$17 + 94 = 111$ ✓
8	18	93	$18 + 93 = 111$ ✓
2	12	99	$\times C = D = 9$
9	19	92	$\times B = C = 9$

- In every case $A = 1$ and $C = 9$; only B and D move, and they move together since $D = 8 - (B - 3)$.

2.N.3 The water jug problem

The eight production rules, state (x, y)

x is the contents of the larger jug, y of the smaller. Written here for the 4-gallon and 3-gallon pair; swap the capacities for any other version. This is the rule set *Insights* and the lecture decks both use, and it is what “write down the production rules” means.

#	Condition	Result	Meaning
1	(x, y) and $x < 4$	$(4, y)$	Fill the 4-gallon jug
2	(x, y) and $y < 3$	$(x, 3)$	Fill the 3-gallon jug
3	(x, y) and $x > 0$	$(0, y)$	Empty the 4-gallon jug on the ground
4	(x, y) and $y > 0$	$(x, 0)$	Empty the 3-gallon jug on the ground
5	(x, y) and $x + y \geq 4$ and $y > 0$	$(4, y - (4 - x))$	Pour from the 3 into the 4 until the 4 is full
6	(x, y) and $x + y \geq 3$ and $x > 0$	$(x - (3 - y), 3)$	Pour from the 4 into the 3 until the 3 is full
7	(x, y) and $x + y \leq 4$ and $y > 0$	$(x + y, 0)$	Pour all of the 3 into the 4
8	(x, y) and $x + y \leq 3$ and $x > 0$	$(0, x + y)$	Pour all of the 4 into the 3

2 gallons in the 4-gallon jug, from a 4 and a 3 4 2+2+4

PIN 2/41 77 Ch 2+2+4, 71 Bh 4 · the classic; *Insights* §2.2.1 works this one

QUESTION AS PRINTED 77 Ch · 2+2+4

Assume you are given two jugs, a 4-gallon one and a 3-gallon one, and a pump which has unlimited water. How can you get exactly 2 gallons of water in the 4-gallon jug? Formalize the problem, write down Production Rules and draw the Search Tree for the water jug problem.

Solution, following *Insights*: rules 2, 7, 2, 5, 3, 7

Step	State (x, y)	Rule	Action
start	$(0, 0)$	—	both jugs empty
1	$(0, 3)$	2	fill the 3-gallon jug
2	$(3, 0)$	7	pour all of it into the 4-gallon jug
3	$(3, 3)$	2	fill the 3-gallon jug again
4	$(4, 2)$	5	pour into the 4 until it is full; the 4 takes 1, leaving 2
5	$(0, 2)$	3	empty the 4-gallon jug
6	$(2, 0)$	7	pour the 2 gallons into the 4-gallon jug

- 6 rule applications, and breadth-first search confirms no shorter path exists.

Formulating it as a state space

- **State:** the ordered pair (x, y) , $0 \leq x \leq 4$, $0 \leq y \leq 3$. **20 states** in all.
- **Initial state:** $(0, 0)$, both jugs empty.
- **Goal state:** $(2, y)$ for any y – the question asks only for 2 gallons *in the 4-gallon jug*, so y is unconstrained.
- **Operators:** the eight rules alongside.
- **Path cost:** 1 per rule application.
- **Why the state is just a pair:** the jugs have no markings, so nothing about *how* the water got there can matter. This is the whole point of the exercise and it is worth one sentence in the answer.
- **Problem type: ignorable** (§2.3) – a wrong move can simply be emptied out, so no backtracking machinery is needed.

Search tree

- From $(0, 0)$ only two rules apply, 1 and 2, so the tree has **two branches at the root**. The 6-step path above is the left branch; the right branch (fill the 4 first) reaches $(2, 3)$ in 6 steps too.
- **Draw it as this, not as a bush** – show the two root branches and expand the one that succeeds:
 $(0, 0)$
 $\rightarrow$ [rule 1] $(4, 0) \rightarrow (1, 3) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (4, 1) \rightarrow (2, 3)$
 $\rightarrow$ [rule 2] $(0, 3) \rightarrow (3, 0) \rightarrow (3, 3) \rightarrow (4, 2) \rightarrow (0, 2) \rightarrow (2, 0)$
- **Both are correct and both are 6 moves.** Either scores full marks; say which you are drawing.
- *Insights* draws this tree with a self-loop child at each level, labelled $(x, y) \rightarrow (x+0, y+0)$. That is not a legal successor – no rule leaves the state unchanged – and it should be left out.

7 gallons in the 9-gallon jug, from a 3, a 5 and a 9 2+1+2+3

PIN 1/41 79 Bh 2+1+2+3 · **the hard variant:** three jugs, and the pump may fill only the smallest

QUESTION AS PRINTED 79 Bh · 2+1+2+3

What are the steps of Problem solving? You are given three empty jugs: a 3-gallon, a 5-gallon and a 9-gallon, and a pump can fill water only in the 3-gallon jug. How do you get exactly 7 gallons in the 9-gallon jug? Formalize the problem, write production rules and draw the search tree.

What makes this one different

- **Only the 3-gallon jug may be filled from the pump.** So rule 1 applies to jug A alone; there is no “fill the 5” and no “fill the 9”.
- Every gallon that reaches the 9-gallon jug arrives in multiples of 3, *unless* the 5-gallon jug is used to strand a remainder. $9 = 3 \times 3$, so a direct route gives only 3, 6 or 9 gallons: **the 5-gallon jug is not optional, it is the whole trick.**
- **State:** the triple (a, b, c) for the 3, 5 and 9-gallon jugs. **Initial** $(0, 0, 0)$, **goal** $(a, b, 7)$.
- **Operators:** fill A from the pump; empty any jug; pour i into j until i is empty or j is full.

Solution, 9 moves (shortest)

#	(a, b, c)	Action
	$(0, 0, 0)$	start
1	$(3, 0, 0)$	fill A from the pump
2	$(0, 3, 0)$	pour A into B
3	$(3, 3, 0)$	fill A
4	$(1, 5, 0)$	pour A into B; B takes 2 and is full, leaving 1 in A
5	$(0, 5, 1)$	pour A into C
6	$(3, 5, 1)$	fill A
7	$(0, 5, 4)$	pour A into C
8	$(3, 5, 4)$	fill A
9	$(0, 5, 7)$	pour A into C

- **The whole idea is step 4:** fill B to strand exactly 1 gallon in A, tip that 1 into C, then add $3 + 3$ on top.
 $1 + 3 + 3 = 7$.
- B is never emptied after step 4; it has done its job.

1 litre, from a 5-litre and a 2-litre jug 8

PIN 1/41 81 Ash 8 · this paper asks **three things at once:** is it well defined, list the attributes for designing an intelligent agent, and solve it. Answer all three

QUESTION AS PRINTED 81 Ash · 8

You are given two jugs, a 5-litre one and a 2-litre one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. You are required to measure exactly 1 litre of water. Is this a well-defined problem? Justify your argument and list all the attributes to design an intelligent agent. Solve this problem by specifying states and the operations used.

Part 1: is it well defined? **Yes**

- **Initial state** given: both jugs empty, $(0, 0)$.
- **Operators** given: fill from the pump, empty on the ground, pour one into the other.
- **Goal test** given and mechanical: does either jug hold exactly 1 litre?
- **Path cost:** 1 per operation.
- Every clause of §2.2 is satisfied, so it is well defined. It is also **solvable**, since $\gcd(5, 2) = 1$ divides 1.

Part 2: attributes for designing an intelligent agent

- This is asking for **PEAS** plus the environment properties (Ch 1):
 - **P:** measure exactly 1 litre, in the fewest operations, spilling least.
 - **E:** the two jugs, the pump, the ground.
 - **A:** fill, empty, pour.
 - **S:** whether a jug is full, empty, or neither.
- Environment: **fully observable, deterministic, sequential, static, discrete, single agent.** A **goal-based agent** is the right type, because the answer needs a *sequence* planned ahead, not a reflex.

Part 3: solution, 4 moves (shortest)

#	(x, y)	Rule	Action
	$(0, 0)$	–	start
1	$(5, 0)$	1	fill the 5-litre jug
2	$(3, 2)$	6	pour into the 2-litre jug until full
3	$(3, 0)$	4	empty the 2-litre jug
4	$(1, 2)$	6	pour again; the 2 takes 2 more, leaving 1 in the 5

- **The 1 litre ends up in the 5-litre jug**, which is what the question asks for: it says “measure exactly 1 litre”, not “in the 2-litre jug”.
- State space: $6 \times 3 = 18$ states.
- **The general rule worth one extra mark:** with jugs of capacity m and n , exactly the multiples of $\gcd(m, n)$ up to $\max(m, n)$ are measurable. Here $\gcd(5, 2) = 1$, so every whole number of litres from 0 to 5 can be measured.

2.N.4 Farmer, wolf, goat and cabbage

Formulate the river crossing as search and solve it 8

PIN 1/41 73 Ma 8 · the standard example of a problem whose interest is entirely in the **safety precondition**

QUESTION AS PRINTED 73 Ma · 8

A farmer has a goat, a wolf and a cabbage on the west side of a river and wants to move everything to the east side. The boat holds the farmer and one other thing. The wolf eats the goat and the goat eats the cabbage if left alone. Formulate this puzzle as search and solve it (any method); draw the search tree and final solution.

Formulation

- **State:** the 4-tuple (F, W, G, C) giving which bank the Farmer, Wolf, Goat and Cabbage are on. Write 0 for west, 1 for east.
- **Initial state** $(0, 0, 0, 0)$; **goal state** $(1, 1, 1, 1)$.
- **Operators:** the farmer crosses, taking nothing or exactly one item that is currently on his own bank. **Four operators.**
- **Safety constraint** (this is what makes it a puzzle): a state is illegal if
 - the goat and the wolf are together without the farmer, or
 - the goat and the cabbage are together without the farmer.
- **Path cost:** 1 per crossing.
- **State space:** $2^4 = 16$ states in total, of which **only 10 are safe and reachable**. Six are pruned by the constraint, and that pruning is the answer to “how does a constraint reduce the search”.

Solution: 7 crossings

#	(F, W, G, C)	Crossing
	W W W W	start, everything on the west
1	E W E W	take the goat across
2	W W E W	return alone
3	E E E W	take the wolf across
4	W E W W	bring the goat back
5	E E W E	take the cabbage across
6	W E W E	return alone
7	E E E E	take the goat across

- **Step 4 is the step everyone misses.** Nothing else works: after the wolf is delivered, the goat cannot stay with it, so it must come back and wait while the cabbage goes over.
- The goat is the only item that is dangerous *in both directions*, which is why it moves first, and why it is the one that gets brought back.
- **7 is the minimum.** There is a mirror-image solution that takes the cabbage third instead of the wolf; it is also 7 crossings and equally correct.

2.N.5 Tic-tac-toe as a state space

Describe tic-tac-toe in terms of state space, data structure and goal 7

PIN 1/41 72 Ma 7 · the paper asks for **complete state space, efficient data structures and goal**, i.e. three named parts. Answer in three parts

QUESTION AS PRINTED 72 Ma · 7

Consider a Tic-Tac-Toe game playing as a problem. Prepare a brief description of your problem in terms of complete state-space, efficient data structures and goal, which are to be used for a machine to play that game.

1. Complete state space

- A state is an assignment of **blank**, **X** or **O** to each of the 9 squares, plus whose turn it is.
- **Naive count:** $3^9 = 19,683$ boards. **Most are not reachable** – they have five O's and no X's, or two completed lines.
- **Reachable, legal count:** **5,478** states, counting every board that can arise in a real game with X moving first and play stopping at a win.
- **Branching factor** 9 at the root, falling by one each ply; at most $9! = 362,880$ complete play sequences, which is small enough to search **exhaustively**. That is the point worth making: tic-tac-toe is one of the few games where minimax can see to the end of the game.
- **Initial state:** the empty board. **Operators:** place your mark on any blank square. **Problem type:** **irrecoverable** (§2.3) – a placed mark cannot be lifted, so the machine must plan before moving.

2. Efficient data structure

- **A 9-element vector** indexed 1–9, holding 0 for blank, 1 for X, 2 for O. Simple, and a move is one array write.
- **The efficient version:** two **9-bit masks**, one for X and one for O. Then
 - a legal-move test is a bitwise AND against the union,
 - a win test is `mask & line == line` over the **8 winning lines** (3 rows, 3 columns, 2 diagonals), which is 8 AND operations and no loops,
 - the whole board fits in one integer, so states can be hashed and cached.
- A move-indexed **lookup table** of best replies is the fastest of all, but it is the table-driven agent of Ch 1: correct, and impractical to build by hand.

3. Goal

- **Goal test:** any of the 8 lines is filled with one player's mark, or all 9 squares are taken (a draw).
- **Utility:** +1 win, 0 draw, –1 loss, which is what minimax (§3.4) needs.
- **The honest closing line:** with both players optimal, tic-tac-toe is a **forced draw**. The machine's real goal is therefore *never to lose*, and to win only when the opponent errs.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
AI	artificial intelligence	ANI	artificial narrow intelligence
AGI	artificial general intelligence	ASI	artificial super intelligence
KR	knowledge representation	KRR	knowledge representation and reasoning
PEAS	performance measure, environment, actuators, sensors	GPS	General Problem Solver
NLP	natural language processing	NLU	natural language understanding
NLG	natural language generation	CAPTCHA	completely automated public Turing test to tell computers and humans apart
ML	machine learning	ANN, NN	artificial neural network
MLP	multilayer perceptron	BP	back-propagation
SOM	self-organizing map	GA	genetic algorithm
FIS	fuzzy inference system	ID3	Iterative Dichotomiser 3
CSP	constraint satisfaction problem	DFS	depth first search
BFS	breadth first search	DLS	depth limited search
IDS	iterative deepening search	UCS	uniform cost search
CNF	conjunctive normal form	WFF	well-formed formula
FOL	first-order logic	FOPL	first-order predicate logic
RRS	resolution refutation system	BBN	Bayesian belief network

Short	Means	Short	Means
ES	expert system	KB	knowledge base
CD	conceptual dependency	LISP	list processing (the language)
attr(s)	attribute(s)	repo	repository
freq	frequency	config	configuration
info	information	no.	number
Eg	example	Vs	compare / differentiate
w/, w/o	with, without	+Fig	draw the figure
→	leads to, therefore	=	is, equals

Reading the year tags and chips

Exam-paper month codes: Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

Bold year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

Three course codes, three faces. Plain = **CT 653** (BCT, III-II, 26 papers) · *monospace* = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

Tier chips over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth ≥ 6 marks.

Mark chips such as **2+6** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic never yet examined but written up anyway. **verified/added** marks anything researched and added that no source note in this folder covers.